Lecture 1

Units and Measurements



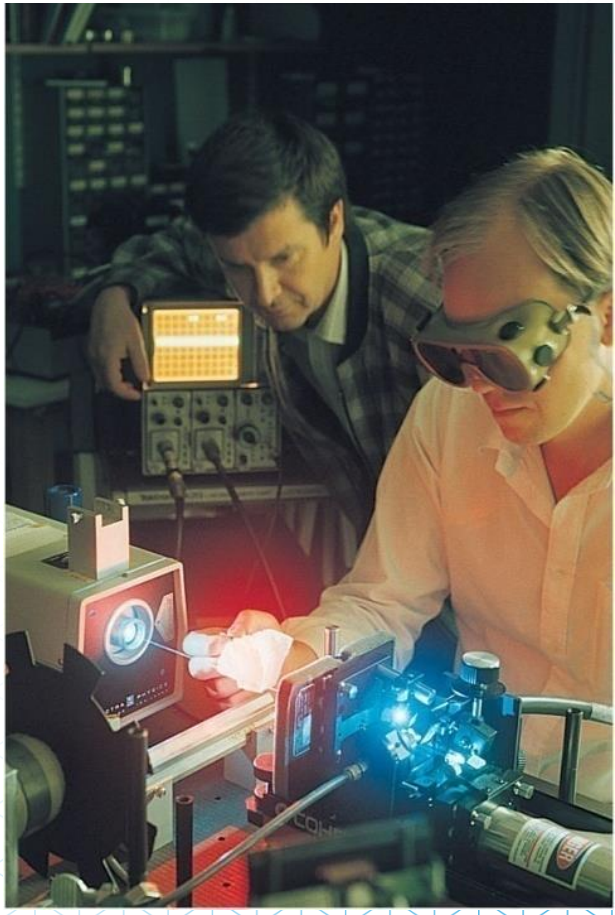
Outlines

- **What is physics?**
- **Physical Quantities**
  - Fundamental Quantities
  - Derived Quantities
  - System of Units
  - Prefixes
- **Conversion of Units**

What is Physics?

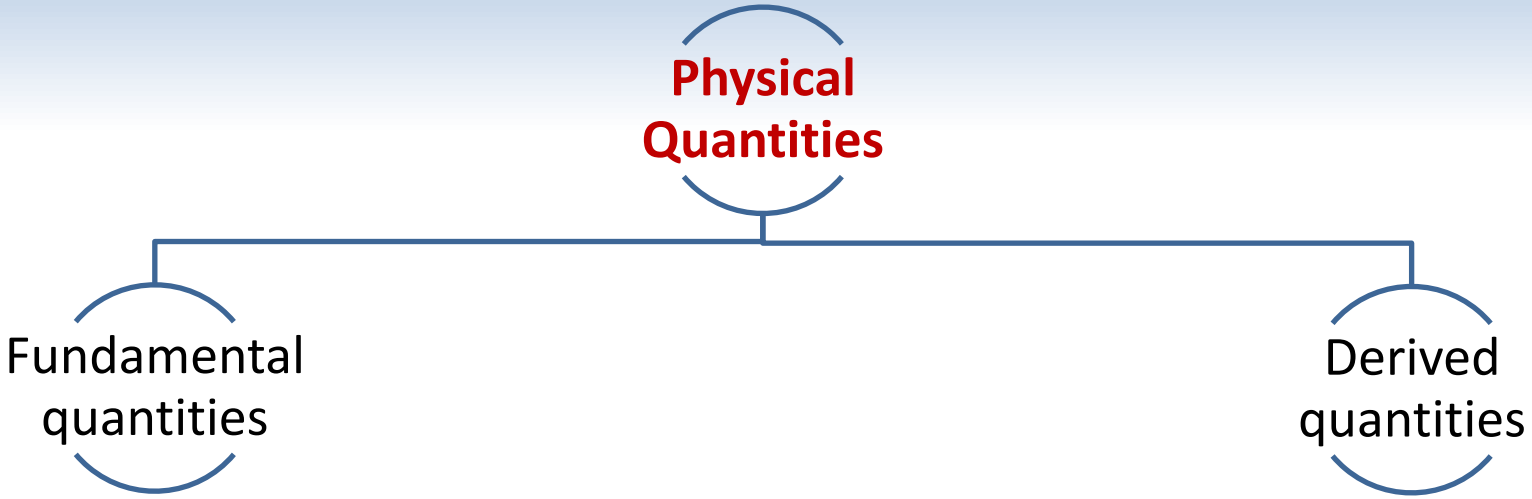
- **Physics** is the foundational science that explores the fundamental laws of nature, governing everything from the smallest subatomic particles to the vast structures of the universe.

- Science and engineering are based on measurements and comparisons. Thus, we need rules about how quantities are measured and compared, and we need experiments to establish the units for those measurements and comparisons.



Physical Quantites

- A **Physical Quantity** is any **physical property** that can be **measured.**



1) Length
2) Mass
3) Time
4) Temperature
5) Electric current
6) Luminous intensity
7) Amount of Substance

- **Those are that can be expressed in terms of fundamental quantities**
  Area
- Volume
- Velocity
- Acceleration
- Force
- ……

System of Units

- Physical quantities measured by using unit.
  Example: Length is a physical quantity that is measured in unit of meter.

- There are three systems of units:

Quantity	SI or MKS system	French or GCS system	British system
Length	Meter (m)	Centimeter (cm)	Foot (ft)
Mass	Kilogram (kg)	Gram (g)	Pound (lb)
Time	Second (s)	Second (s)	Second (s)
Temperature	Kelvin (°K)	Kelvin (°K)	Fahrenheit (°F)
Electric Current	Ampere (A)	Ampere (A)	Ampere (A)

- In 1960, General Conference on Weights and Measures decided on a universal system of unit called the International System or SI of units.
- SI units is the standard units we use.
- [illegible]

Quantity	Unit of Measurements	Symbol
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	°K
Electric Current	Ampere	A
Amount of substance	Mole	mol
Luminous intensity	Candela	cd

➢ **Example 1:** The Empire State Building is 1472 ft high. Express this height in both meters and centimeters. (hint: 1 ft = 0.3048 m = 30.48 cm)

- Solution: To do the first unit conversion (feet to meters), we can use the relation: 1 ft = 0.3048 *m*

- We set up the conversion factor so that:

$$1472\ ft = 1472\ (0.3048\ m) = 448.67\ m.$$

- So, the height can be expressed as 448.67 m.

- To convert this to centimeters, we use: 1 m = 100 *cm*

$$448.67\ m = 448.67\ (100cm) = 4.4867 \times 10^4\ \text{cm}.$$

- The Empire State Building is 4.4867×10^4 cm high!

- **Example 2:** A rectangular building lot is 100.0 ft by 150.0 ft. Determine the area of this lot in square meters (m^2). (hint: 1 ft = 0.3048 m = 30.48 cm)

- Solution: The area of a rectangle is the product of its length and width, so the area of the lot is:

$$A = (100.0\ \text{ft}) \times (150.0\ \text{ft}) = 1.500 \times 10^4\ \text{ft}^2$$

- To convert this to units of m^2, we use the relation: 1 ft = 0.3048 *m*

- Since we are converting square feet to square meters, so we need to apply the conversion twice:

$$1.500 \times 10^4\ \text{ft}^2 = 1.500 \times 10^4\ (0.3048^2\ \text{m}^2) = 1.393 \times 10^3\ \text{m}^2$$

- Thus, the area of the lot is $1.393 \times 10^3\ \text{m}^2$.

Prefixes

- When working with physical quantities, it's often necessary to express very large or very small numbers.

- Prefixes simplify these numbers by scaling them with powers of ten. Understanding these prefixes is crucial for engineers, as they encounter measurements across a vast range of scales, from nanoscale components to large structural systems.

Power	Prefix	Abbreviation	Power	Prefix	Abbreviation
10^{-18}	atto	a	10^{3}	kilo	k
10^{-15}	femto	f	10^{6}	mega	M
10^{-12}	pico	p	10^{9}	giga	G
10^{-9}	nano	n	10^{12}	tera	T
10^{-6}	micro	μ	10^{15}	peta	P
10^{-3}	milli	m	10^{18}	exa	E
10^{-2}	centi	c			
10^{-1}	deci	d			

- Using these prefixes makes it easier to represent measurements compactly.
- Prefixes correspond to powers of 10.
- Each prefix has a specific name.
- Each prefix has a specific abbreviation.
- The prefixes can be used with any basic units.
  For example:
  - A distance of 0.000001 meters can be written as 1 µm (micrometer).
  - A data storage capacity of 1,000,000,000 bytes is written as 1 GB (gigabyte).
  - 10^{-3} m = 1 mm
  - 10^{3} g = 1 kg

Conversion of Units

- Any quantity can be measured in several different units.
  Example: 20×10^{-3} km = 20 m = 20 x 100 cm = 20 x 1000 mm

- **Problem 1:** A human hair is approximately 50 µm in diameter. Express this diameter in meters.

  Solution: diameter = 50 µm

  Since $\mu = 10^{-6}$, then the diameter = 50×10^{-6} m

- **Problem 2:** if a radio wave has a period of 1 µs, what is the wave's period in millisecond?

  Solution: period = 1 µs

  Since $\mu = 10^{-6}$, then the period = 1×10^{-6} s

  Now we convert to millisecond as, period = 1×10^{-3} ms